

Supplementary Material for Budgeted Placement of Executable Validation Policies on Data Pipeline Lineage

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ABSTRACT

This supplement carries the material the main paper delegates. It states and proves the planner bound with its first-effective condition and reports the planner scaling measurements. It gives the rule-derived policy maps, the M1 propagation metrics, and the complete measurements of the M3 disposition ladder, the M4 placement study, and the M5 order-fork composition probes. It records the sealed protocols of all three fresh evaluations, covering campaign rosters, the complete E1 evaluation figure, the budget grid, the re-measured cost catalog with its component breakdown and cost-damage view, and the launch and promotion criteria. It derives the E2 dose-response prediction and tabulates its outcome. It tabulates the near-band stress runs, the forced false-positive sweep, the signal comparison against the testbed’s own shipped dbt test suite, and the damage-metric sensitivity check. It documents the TPC-DI port, its response classes, the temporal snapshot split, the roster variants the precheck refused, and the exploratory measurement of one of them, and it closes with a guide to the artifact. All values are measured, and the evidence files they are computed from are named throughout.

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PVLDB Artifact Availability:

The source code, data, and/or other artifacts have been made available at <https://github.com/llfuture/lineageguard>.

1 RULE-DERIVED POLICY MAPS

Per node and signal shape, the derivation rule deploys the disposition with the lowest development-measured residual and defaults to no-op unless the action strictly reduces damage. The derived map for the testbed is Table 1.

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Table 1: Rule-derived policy maps on the testbed. Empty cells are no-op. Every entry comes from the development singleton measurements, with no hand-chosen entries.

node	duplicate	numeric	null	fk
stg_products	dedup	quarantine	–	–
products	dedup	quarantine	quarantine	–
order_items	–	–	–	–
orders	dedup	–	–	–
customers	–	quarantine	–	–

Two entries deserve comment. At orders, numeric maps to no-op although the signal fires there, because the measured singleton residual is 1.0, and a mutating action with no benefit is never deployed since it can destroy downstream signal, as §7 shows. At order_items, every measured response is a no-op for every sink-visible family, so the node is never selected and its 52.4s cost is discarded by cost domination.

2 PLANNER BOUND, PROOF, AND SCALING

The planner first collapses nodes with identical measured response signatures to their cheapest representative, reusing the identifiability gate’s equivalence classes, and then runs branch-and-bound in cost order with the per-family bound stated here.

Bound. For a partial plan with remaining affordable node set U and committed set A , let

$$LB = \frac{1}{|F|} \sum_{f \in F} \min_{v \in U \cup A} R_f(v), \quad (1)$$

with F the family set and $R_f(v)$ the measured response of node v on family f .

Condition, first-effective responses. For every family f and every feasible completion π , the plan’s response $R_f(\pi)$ equals $R_f(v)$ for some single $v \in \pi$, namely the first node in DAG order whose deployed rule can fire on f with a state-mutating disposition.

Claim. Under the condition, $LB \leq R(\pi)$ for every feasible completion π , so pruning on LB cannot discard an optimal plan.

Proof. Each per-family term of $R(\pi)$ is $R_f(v)$ for some v the completion draws from $U \cup A$, and a minimum over all of

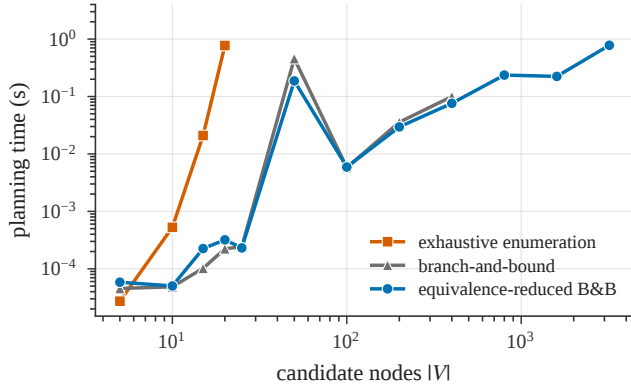


Figure 1: Planning time against node count. Exhaustive enumeration is infeasible past 25 nodes, while equivalence-reduced branch-and-bound stays sub-second to 3,200 nodes and matches the exhaustive optimum wherever both run.

Table 2: Planning time in seconds with 8 families and 8 response classes, medians over deterministic seeded instances. BnB is branch-and-bound, and eq+BnB adds equivalence reduction. Dashes mark infeasible or not-run settings. The equivalence-reduced search matches the exhaustive optimum on every instance where both run. Computed from planner-scaling.json.

$ V $	exhaustive	BnB	eq+BnB
10	0.0005	0.00005	0.00005
20	0.77	0.0002	0.0003
50	—	0.46	0.19
200	—	0.036	0.030
800	—	—	0.24
3200	—	—	0.78

$U \cup A$ cannot exceed the particular element chosen. Averaging over F gives the claim.

The condition is exactly what the main paper’s eligibility rule enforces and what the state-inertness probes of §7 test. Inert nodes drop out of the effective signature, so a plan containing several nodes still presents one effective node per family. The statement is per-family and deliberately claims nothing per campaign. Multi-family campaigns do not decompose, because damages overlap at shared sinks. On ord-mixed, dedup at orders measures 0.4898 and the customers-only plan 0.7614, while their composition measures 0.2469, below the 0.2543 that averaging the families would predict. Applying the bound across a mixed campaign would therefore over-estimate the residual and could prune the optimum, so the planner scores mixed campaigns as their own scenario from the measured composed signature and lets only single-family scenarios enter the bound. The bound only prunes. Plan responses are always the measured effective signature, which is why the static-plan pilot’s min-envelope failed.

Table 3: M1 macro results on 30 fresh validation campaigns, with a paired, family-stratified, campaign-level bootstrap of 10^4 resamples, whose 95% CIs are [99.72, 99.88] for Brier and [63.62, 82.36] for weighted MAE. ECE is expected calibration error, allowed to degrade by at most 0.02. Computed from the rq1-w30 evaluation.json.

Metric	topology null	calibrated	improvement
Brier	0.087289	0.000166	99.81%
log loss	0.247636	0.006881	97.22%
weighted MAE	0.000496	0.000136	72.67%
ECE	≈ 0	0.00672	within 0.02 cap

With the exact evaluation at every search node, the search returns an optimal plan. Exhaustive enumeration is $O(2^{|V|})$ and infeasible past roughly 25 nodes. Planning time is non-monotone in $|V|$, as Table 2 shows. The 50-node instances are the hardest we generated, because their response classes and budget grid leave the bound least able to prune, while larger instances with more redundant nodes collapse further under equivalence reduction before the search begins.

3 M1 PROPAGATION METRICS

Table 3 gives the complete M1 comparison between the topology null and the calibrated model on the 30 fresh validation campaigns, behind the Brier and weighted-MAE headlines the main paper quotes.

4 MODEL-FITTING DETAILS

The calibration scalars a and b of the propagation model are fit by campaign-equal binary cross-entropy with exact Newton and a ridge on b alone. The response model fits, per action, an exact-arithmetic least-squares model of the surrogate’s error on the three factorial factors, with rational coefficients, no interactions, and leave-one-cell-out cross-fitting, and predicts $\max(0, \text{surrogate} + \text{residual})$.

5 M3 DISPOSITION LADDER, COMPLETE TABLE

Every entry is an exact rational. In lowest terms the macro column reads 1, 31/38, 1/2, 2, 0, 29/38, 5/19, 1/2. The numeric column is 10/19 wherever quarantine fires, and the duplicate column is 21/19 under legacy quarantine and exactly 0 under dedup. The main text quotes these as decimals. Fail-closed reaches zero damage, loses availability entirely, and is inadmissible under the declared caps. Column null-out is harmful under exact-row damage, since a NULL is as changed as a wrong value and blanking both copies of a duplicate adds mismatching rows.

6 M4 PLACEMENT RESPONSES

Figure 2 gives the per-node responses and Table 5 the resulting placement spread per policy over 80 placements and 160 branch executions plus eight anchors. The two upstream

Table 4: M3 responses on the development cells, 64 policy-cell measurements over 120 physical branch executions with zero technical failures. The macro, num., and dup. columns give NRD overall and per family. SESOI marks whether the policy’s improvement over legacy quarantine reaches the pre-registered 10%, and safe means within the clean-collateral and availability caps. Computed from d9-mve-summary.json.

policy	macro	num.	dup.	SESOI	safe
no validation	1.0000	1.0000	1.0000	–	yes
quarantine (legacy)	0.8158	0.5263	1.1053	no	yes
dedup only	0.5000	1.0000	0.0000	yes	yes
null-out only	2.0000	1.0000	3.0000	no	yes
fail-closed	0.0000	0.0000	0.0000	yes	no
conditional quar./no-op	0.7632	0.5263	1.0000	no	yes
conditional quar./dedup	0.2632	0.5263	0.0000	yes	yes
conditional null-out/dedup	0.5000	1.0000	0.0000	yes	yes

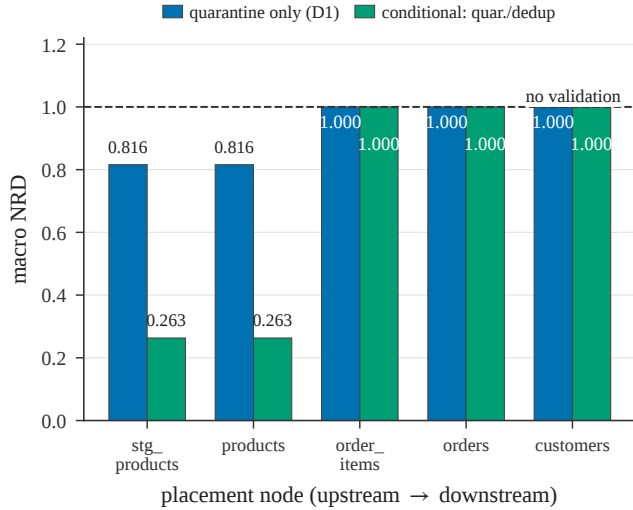


Figure 2: The same policy placed at each of five nodes along the lineage chain. Under quarantine only, moving from the best node to a downstream one costs 0.1842 macro NRD, and under the conditional quarantine and dedup policy the same move costs 0.7368. How much placement matters depends on the disposition space.

nodes produce identical responses, and the three mart-level nodes leave scored damage unchanged for products-side errors because they cannot repair outputs on another branch or earlier in the lineage.

Table 5: M4 placement spread per policy, the difference between worst and best macro NRD over the five chain nodes. Computed from d10-summary.json.

policy	best / worst placement	spread
quarantine only	0.8158 / 1.0000	0.184
conditional quar./dedup	0.2632 / 1.0000	0.737

Table 6: M5 measurements on the train snapshot, 34 rows with zero technical failures and zero clean collateral, k injected rows per cell. Computed from d11-merged.json.

cell	action	NRD
ord-dup ($k=100$)	dedup@orders	0.0000
	quarantine@orders	1.0000
	quarantine@orders	1.0000
ord-num ($k=100$)	null-out@orders	1.0000
	quarantine@customers	0.5086
	quarantine@products	0.5263
null-prod ($k=1$)	quarantine@orders	1.0000
	quarantine@orders	1.0000
fk-ord ($k=100$)	conditional@orders	0.4898
	static dedup@orders	0.4898
	static quarantine@orders	1.0000
	customers only	0.7614
Composition probes		
ord-num	quar.@orders → quar.@customers	1.0000
ord-num	derived, inert@orders + customers	0.5086
ord-mixed	derived, dedup@orders + quar.@customers	0.2469
ord-mixed	static set with same effective signature	0.2469

7 M5 ORDER-FORK RESPONSES AND COMPOSITION PROBES

The first probe is the composition hazard. Quarantine at orders is useless in isolation at 1.0, yet it destroys the customers-side signal when executed upstream of the beneficial customers action, turning 0.5086 into 1.0. The remaining probes confirm the state-inertness rule. Replacing the mutating no-win action by no-op restores the downstream response exactly, and two plans with the same effective signature measure identically.

8 E1 SEALED PROTOCOL OF THE FRESH EVALUATION

Figure 3 is the complete three-panel view of E1, whose per-family panel the main paper carries.

8.1 Campaign roster

The 16 fresh campaigns are materialized on the validation snapshot, which holds 1.77M order_items, 1.22M orders, and 3,102 customers rows, strictly after the plans were sealed. Targets are derived by domain-separated SHA-256 ranking, so no random-number generator state is involved anywhere.

The products-fork campaigns use the four lowest- and four highest-fan-out products, with fan-out measured on the clean validation snapshot, one product per campaign and no reuse. Deletion and FK campaigns are negative controls, since no deployed rule can fire on deleted rows and the orphaned

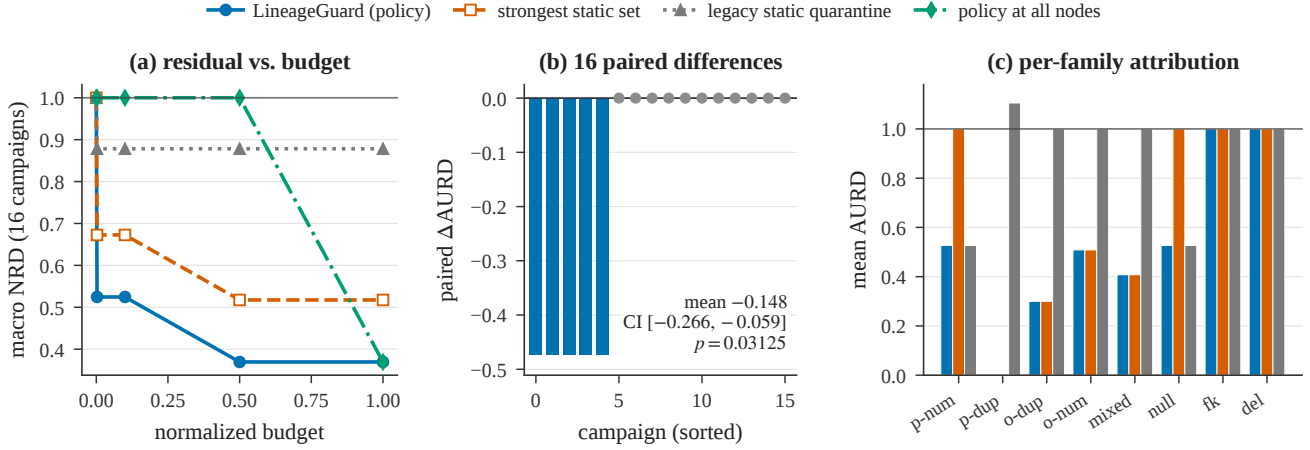


Figure 3: The E1 fresh paired policy evaluation over 16 campaigns and six error families. Left, macro NRD against normalized budget. Center, per-campaign paired AURD differences between policy and strongest static. Right, per-family mean AURD attribution, including the two no-win families. The right panel also appears in the main paper.

Table 7: Fresh campaign roster, where k is the number of injected rows.

family	campaigns	injection	k
prod-num	4	price + 10^4 cents (src) / +\$100 (stg)	1
prod-dup	4	duplicate product row (src / stg)	1
ord-dup	2	duplicate order rows	1, 200
ord-num	2	subtotal and total + 10^7 cents	1, 200
ord-mixed	1	50 numeric + 50 duplicate, disjoint rows	100
null-prod	1	product price set to NULL	1
fk-ord	1	customer key replaced by orphan value	200
del-ord	1	order rows deleted	200

orders lose their true customer identity, so no disposition can restore the clean state.

8.2 Budget grid and the re-measured cost catalog

The budget grid derives from the re-measured full-policy cost catalog, summing deploy, detect, and worst-case disposition per node. The per-node costs in μ s are 82,942 at stg.products, 85,132 at products, 97,882 at customers, 47,324,862 at orders, and 52,351,868 at order.items, for a total of 99,942,686. The grid points are 0, then 265,956 covering the three cheap nodes, then 10%, then 50% where the orders node becomes affordable, and 100%. The earlier deployment-only catalog

Table 8: Re-measured policy costs in μ s, the worst-case per-workload medians from five single-threaded trials after one warm-up across nine workloads. C_{deploy} , C_{detect} , and $C_{disp}^{\max}$ are the deployment, detection, and worst-case disposition components, and the under column is the factor by which the deployment-only catalog understated the full policy cost. Computed from policy-cost-catalog.json.

node	C_{deploy}	C_{detect}	$C_{disp}^{\max}$	under
products	4,708	77,996	2,428	18.1×
stg_products	6,813	73,778	2,351	12.2×
customers	28,211	69,671	0	3.5×
order_items	51,078,023	967,902	305,943	1.02×
orders	47,159,369	165,493	0	1.00×

understated the cheap nodes by up to 18.1 $\times$, as Table 8 shows, which is why the grid was re-derived before this evaluation.

Each component is measured single-threaded with one warm-up plus five trials over nine workloads, taking per-workload medians and the worst case over workloads, with a transaction rollback between trials. The five-node total changes only from 98.28s to 99.94s under the corrected catalog, because the expensive nodes dominate the sum. Dispositions themselves are cheap, at 0.8 to 2.4ms on the mart nodes, except exact dedup on the 1.05M-row order.items relation of the train snapshot at 306ms. Costs were measured with the machine otherwise idle. The M3 and M4 damage runs were sharded four ways, which is sound for deterministic damage and makes any timing recorded during them unusable.

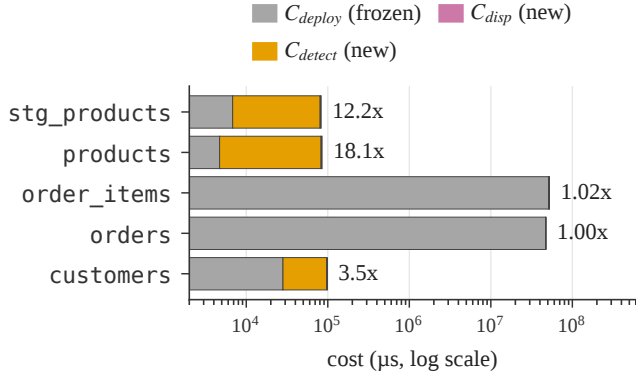


Figure 4: Re-measured policy cost components. At the cheap nodes detection dominates, and the deployment-only catalog understates true policy cost by up to 18 $\times$.

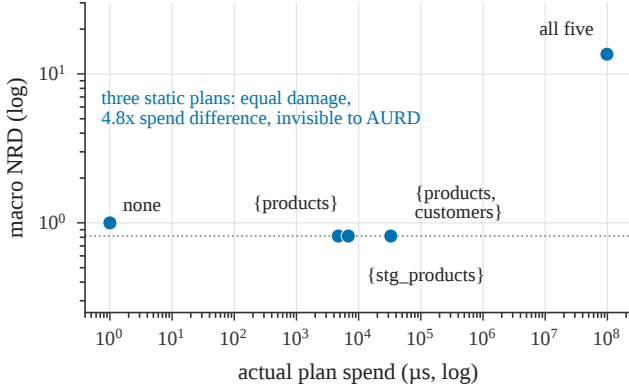


Figure 5: Cost-damage behavior of all physically executed plans in the static-plan pilot. Among three plans with identical damage, the cheapest is Pareto-efficient while the other two spend up to 4.8 $\times$ more for the same result.

8.3 The cost-damage view of E4

Figure 5 plots each physically executed plan of the static-plan pilot by its measured spend and resulting damage. Three plans have identical damage and costs differing by up to 4.8 $\times$, and only the cheapest is Pareto-efficient. The pilot’s primary sat in the dominated set, which is why cost domination became a planner rule in the main paper.

8.4 Launch criteria of the identifiability gate

The evaluation may launch only if five conditions hold, computed from development data alone. C1 requires the primary and comparator plans to be response-distinguishable on at least one campaign family. C2 requires the predicted relative AURD improvement to reach the pre-registered 10%. C3

Table 9: Development responses supporting E2, per split. Here w is the numeric share of the mixed no-validation damage, overlap is $(D_{\text{num}} + D_{\text{dup}} - D_{\text{mix}}) / D_{\text{mix}}$, and $q(k)$ is the implied quarantine ratio, matching $N / (2N - k)$ at $N=10$ exactly at every measured k . The cond, dedup, and quar columns give the macro NRD of the conditional policy, static dedup, and static quarantine.

split	w	overlap	cond	dedup	quar	$q(k)$
1 $\times$ 9	0.2636	+8.2%	0.1388	0.2636	2.636	10/19
3 $\times$ 7	0.6231	+16.2%	0.3665	0.6231	2.077	10/17
5 $\times$ 5	0.8333	+16.7%	0.5556	0.8333	1.667	10/15
7 $\times$ 3	0.9471	+12.4%	0.7285	0.9471	1.353	10/13
9 $\times$ 1	0.9947	+4.7%	0.9043	0.9947	1.105	10/11

forbids any selected plan from relying on an unmeasured composition. C4 forbids a cost-dominated primary. C5 requires that no two measurements sharing an effective signature disagree. All five held with a predicted improvement of 26.21%, and the decision was LAUNCH.

8.5 Promotion criteria of the one-shot gate

The gate recomputes ten criteria from raw measurement rows. G1 requires 16 of 16 campaigns complete. G2 requires all method-budget-campaign bindings to resolve. G3 requires zero technical failures. G4 requires all no-validation damages strictly positive. G5 requires zero clean collateral and full availability on all primary and comparator placements. G6 requires the primary and comparator to be physically distinguishable. G7 requires a negative mean paired AURD difference. G8 requires a negative bootstrap 95% upper bound. G9 requires a relative improvement of at least 10%. G10 requires an exact one-sided sign-flip $p \leq 0.05$. All ten held, and the decision was GO. The statistics are a paired campaign percentile bootstrap with 10,000 resamples and a recorded fixed seed, an exact sign-flip over all 2^{16} assignments, and Holm correction over the two secondary comparisons.

9 E2 PROTOCOL, DEVELOPMENT MEASUREMENTS, AND DOSE-RESPONSE

9.1 Development measurements for E2

This train-role study comprises 34 measurement rows over 52 branch executions, with no technical failures and no clean collateral. It measured products-fork mixed cells, in which k_{num} numeric and k_{dup} duplicate errors land on disjoint SKUs with one ranking per cell split by position, plus one $k=4$ pure-numeric cell. Component anchors of numeric-only and duplicate-only injections measure the numeric damage share $w = D_{\text{num}} / D_{\text{mixed}}$ per split. Computed from d12-measurement.json.

Static dedup equals w exactly on all five splits, so dedup restores the duplicate component to the numeric-only state. The equivalence-class probe measured identical responses for

the conditional policy at stg_products and at products, both exactly 5/9, and the $k=4$ numeric cell measured $0.625 = 10/16$, confirming $q(k)$ off the mixed family.

9.2 Sealed protocol and gate outcomes

Eighteen campaigns run on the validation snapshot, five mixed splits as in Table 9, four $k=1$ prod-numeric, one $k=4$, two null, two prod-dup, ord-dup and ord-num at $k=20$, and fk and deletion at $k=200$, under the same budget grid, SESOI, seeds, and statistics as E1 with the exact sign-flip enumerated over 2^{18} assignments. The raw_products table has 10 SKUs, so SKU reuse across rounds is unavoidable and recorded, while within-campaign numeric and duplicate targets are disjoint by construction. Launch criteria C1 to C5 all held with a predicted relative improvement of 31.02% and decision LAUNCH, and promotion criteria G1 to G10 all held with decision GO, the aggregator and gate agreeing to 10^{-9} . Observed values are mean AURD 0.5160 for the policy against 0.7483 for the strongest static plan, a relative improvement of 31.03%, a bootstrap 95% CI of $[-0.325, -0.139]$, $p = 2^{-12}$, 12 of 18 non-zero paired differences with 7 distinct values, zero failures, and zero clean collateral with full availability on all 108 placements. Computed from p3-protocol.json, p3-precheck-result.json, p3-summary.json, and p3-gate-result.json.

9.3 The pre-registered dose-response

The statement sealed before launch reads as follows. Each mixed campaign’s paired difference equals $(q(k_{\text{num}}) - 1) w f_{\text{grid}}$ with $q(k) = N/(2N - k)$ at $N=10$ and $f_{\text{grid}} = 1 - x_1/2$, where x_1 is the first non-zero normalized budget point and w is measured on the fresh campaign’s own plan-free anchor runs, recorded in p3-dose-anchors.json.

The grid factor is derived, and we give its two lines. At budget 0 no plan is feasible and the paired difference is 0, and from x_1 onward both plans hold their selections, so the difference is the constant $\Delta_0 = (q(k_{\text{num}}) - 1) w$. Integrating that step by the trapezoid rule over the normalized grid gives

$$\Delta \text{AURD} = \underbrace{\frac{1}{2}(0 + \Delta_0) x_1}_{[0, x_1]} + \underbrace{\Delta_0 (1 - x_1)}_{[x_1, 1]} = \Delta_0 \left(1 - \frac{x_1}{2}\right),$$

which is $f_{\text{grid}} = 1 - x_1/2$. With the shared E1 and E2 grid, $x_1 = 2.66 \times 10^{-3}$ and $f_{\text{grid}} = 0.9986695$. Every factor on the right-hand side is therefore either derived from Eq. 1 of the main paper or measured on the campaign’s own anchors, and none is fitted to the paired differences the prediction is tested against.

10 E3 STRESS MEASUREMENTS

10.1 Near-band magnitude sweep

On the orders fork detection turns on once the per-row value crosses the band, but the residual stays 1.0 because no disposition at orders helps that family, as Table 6 shows. The detection curve and the effect curve are separate properties. Band drift is real. The clean validation maximum of customers.lifetime.spend exceeds the train maximum by 41%, at

Table 10: E2 dose-response, observed paired AURD differences against the pre-registered line. Computed from p3-dose-analysis.json.

split	w (fresh)	predicted	observed	residual
1×9	0.2636	−0.1247	−0.1247	-1.4×10^{-17}
3×7	0.6231	−0.2562	−0.2562	-1.1×10^{-16}
5×5	0.8333	−0.2774	−0.2774	0
7×3	0.9471	−0.2183	−0.2183	-5.6×10^{-17}
9×1	0.9947	−0.0903	−0.0903	-2.8×10^{-17}

Table 11: Near-band stress runs on the train side, 7 runs with 0 failures and 0 clean-side fires. The policy is the derived conditional policy at the fork’s acting node. Computed from rqf-stress-measurement.json.

fork	injected delta	detected	NRD
orders	+\$20 (0.6× band edge)	no	1.0000
orders	+\$110	no	1.0000
orders	+\$250	yes	1.0000
orders	+\$100k	yes	1.0000
products	+\$2 (in band)	no	1.0000
products	+\$8	yes	0.5263
products	+\$100	yes	0.5263

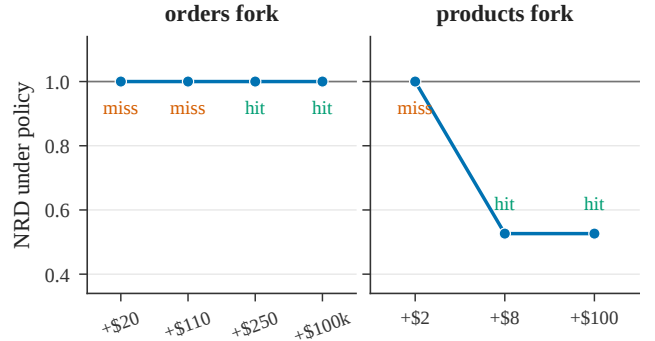


Figure 6: Magnitude sweep, detection and end-to-end effect against injected magnitude under deployed rules. In-band errors are missed and the policy degrades to no-op. Crossing the band restores the full effect on the products fork at 0.5263, while on the orders fork detection turns on but the residual stays 1.0, because no disposition at orders repairs that family. Detection and effect are separate properties. Clean branches show 0 false fires in all runs, with complete values in Table 11.

13,330.72 against 9,454.64, so a $1.2\times$ margin would have false-fired on clean data, while the declared $2\times$ margin produced zero clean fires in all 96 fresh clean branches.

Table 12: The E3 band-margin sweep on TPC-DI, forcing real false positives. The conditional and dedup columns give each policy’s NRD, and the cap column is the clean-collateral admissibility test, computed on the clean branch alone. Computed from b1-fpr-sweep.json.

margin	clean false fires	conditional	dedup	cap
2.0×, 1.0×	0 (0.0%)	0.470	0.596	pass
0.5×	49,095 (49.8%)	1433.5	0.596	violated
0.2×	78,808 (79.9%)	2229.3	0.596	violated
0.05×	93,660 (95.0%)	2669.7	0.596	violated

10.2 Forced false positives on TPC-DI

Table 12 carries the sweep the main paper reads its three findings from. Twenty measurements re-run the TPC-DI conflict campaign with the numeric band tightened below its declared 2.0× margin. Clean-side false fires at stg.financial are 0 at 2.0× and 1.0×, then 49,095 or 49.80% at 0.5×, 78,808 or 79.95% at 0.2×, and 93,660 or 95.01% at 0.05×. The conditional policy’s NRD moves through 0.470, 0.470, 1433.50, 2229.34, and 2669.72, with clean collateral 0, 0, 0.2156, 0.3353, and 0.4015. Static dedup holds at 0.596 with zero collateral at every margin. Static quarantine violates the cap even at the declared margin, at 635.63 with collateral 0.0955, because it also fires on native orphans and nulls. Within the admissible set of zero collateral, the worst case never exceeds 1.0.

10.3 Signal comparison against the shipped dbt suite

The Jaffle Shop project ships 27 data tests and 3 unit tests, authored upstream. The suite uses dbt’s `unique`, `not_null`, and `relationships` assertions plus identity expressions on the order and customer marts, such as `order.total = subtotal + tax.paid`. We inject, rebuild, then run that suite unmodified with failing rows stored, and compare what fires against the deployed predicates of the signal contract on the same branch. The comparison is signals only, since no disposition executes and no damage is measured, in a development role with no gate.

Two rows carry the comparison. On the duplicated product row and the FK orphan the two agree exactly, each flagging the same single row, so the contract’s duplicate and fk channels are the deployed assertions themselves. The numeric rows show the axis on which coverage differs, in both directions. An identity expression fires on a value error whatever its magnitude and stays silent when subtotal and total move together, while a band fires on magnitude and stays silent inside itself. Granularity differs with them, since the contract flags one row at the node where a disposition can act while the identity flags the sink rows that are consequently wrong, which is why the two compose.

Table 13: Signal comparison over eight cases. The contract column is the shape raised by our deployed predicates at the injection node, and the shipped-suite column names the failing tests. Computed from audit-dbt-fidelity.json.

injection	contract	shipped suite
none (clean)	silent	30/30 pass
product price +10 ⁶	numeric	subtotal identity (42,396)
product price +100	silent	subtotal identity (42,396)
product row duplicated	duplicate	unique (1) + down-stream
product price NULL	null	subtotal identity (41,191)
order subtotal and total +10 ⁷	numeric	no failure
order subtotal only +100	silent	4 identity tests
order FK orphan	fk	relationships (1)

11 E5 DAMAGE-METRIC SENSITIVITY

Sixty-two measurements recompute two metric variants inside the same physical executions. Value weighting sums the absolute deviation of sink numeric columns over matched keys, plus the full magnitude of added and deleted rows, normalized by clean magnitude, and the second variant is mean per-row relative error.

The comparison set is defined mechanically so that a reader can reconstruct it. It contains every cell-action pair whose NRD is defined under all three metrics, which is 53 of the 62 rows, the remaining 9 being the no-validation anchors that define the denominators. A pair agrees when the three metrics place it in the same one of three directions, helping at $\text{NRD} < 1$, no-op at NRD within 10^{-6} of 1, or hurting at $\text{NRD} > 1$.

Forty-seven of the 53 agree. The six disagreements arise from two mechanisms. Null-out under magnitude metrics accounts for four. On fin-num at $k=1, 10, 100$ it is an exact-row no-op at 1.000 yet removes almost all value error at 0.004 to 0.005, and on the conflict cell static null-out is harmful under exact-row at 1.385 yet is the best static plan under value weighting at 0.009 and relative error at 0.006. Quarantine on market duplicates accounts for the other two, a no-op under exact-row and value weighting at 1.000 and harmful under relative error at 1.354. Consequently the conditional escape is +21.2% under the declared exact-row metric, −13.2% under value weighting, and −1.0% under relative error. Dedup measures 0.000 under all three metrics wherever it applies. Computed from b2-metric-variants.json.

12 E6 THE TPC-DI SECOND PIPELINE

12.1 Unblocking generation

Three toolchain failures blocked DIGen, all diagnosed and fixed without modifying the pristine tools. The official jar

Table 14: E6 TPC-DI singleton responses in NRD, where dashes mark unmeasured entries. Quarantine on the FK shape costs 1722 because the pipeline carries native orphans that the rule legitimately fires on.

family	node	quar.	dedup	null-out
fin-num ($k=1$)	stg_financial	0.838	1.000	1.000
fin-num ($k=10$)	stg_financial	0.737	–	–
fin-num ($k=100$)	stg_financial	0.795	–	–
fin-num ($k=1$)	fact_financials	0.838	–	–
fin-dup ($k=1$)	stg_financial	1.239	0.000	1.955
fin-dup ($k=100$)	stg_financial	2.447	0.000	3.243
fin-null ($k=10$)	stg_financial	0.805	1.000	–
fin-fk ($k=10$)	stg_financial	1722.0	–	–
mkt-num ($k=10$)	stg_daily_market	1.000	1.000	–
mkt-dup ($k=10$)	stg_daily_market	1.000	0.000	–

spawns PDGF through a hard-coded lowercase Tools/pdgm path, fatal on case-sensitive filesystems and mis-reported upstream as plugin discovery failure. It composes the child Java path as java.home plus /jre/bin/java, requiring the Java-8 layout. And it forwards a closed stdin to the PDGF console as literal null commands, destroying the session. With a provenance-recorded working copy carrying a compatibility symlink, a JDK-8 environment with a jre shim, and an open-but-silent stdin, DIGen completes at scale factor 3 with 476 files and 267 MiB in 41 s.

12.2 Ported pipeline

The load layer materializes all 17 source relations into a DuckDB raw schema, including fixed-width FINWIRE slicing of CMP, SEC, and FIN records per the spec offsets and a streaming parse of the CustomerMgmt XML action log. Eighteen dbt models follow, six staging models covering stg_company at 1,096 rows, stg_security at 1,998, stg_financial at 98,576, stg_daily_market at 1,282,768, stg_trade at 390,978, and stg_account at 9,008, two dimensions, three facts, and seven scored sinks, five aggregate marts and two row-preserving marts, financial_detail at 98,576 rows and security_master at 1,998. Eight candidate action nodes carry the signal contract, stable-key multiplicity, a $2 \times$ clean-max numeric band, NOT NULL contracts, and relationship tests against the parent relation.

12.3 TPC-DI response matrix for E6

The development-role measurements on the train anchor comprise 82 rows over 688 dbt model steps, 598 mechanism shards plus 90 for the ceiling addendum, with zero technical failures. Computed from d8p-shardA.json, d8p-shardB.json, d8p-shardC.json, and d8p-ceiling.json.

Table 15: Placements and response classes per TPC-DI cell. Computed from d8p-shardA/B/C.json plus d8p-ceiling.json, which contributes seven placements to the mixed cell.

cell	placements	classes
fin-num $k=1$ / 10 / 100	7 / 7 / 7	2 / 2 / 2
fin-dup $k=1$ / 10 / 100	6 / 6 / 6	3 / 3 / 3
fin-dup $k=10$ (oracle detector)	1	1
fin-null $k=10$	3	2
fin-fk $k=10$	2	1
fin-mixed 20×20	14	6
mkt-num $k=10$	4	1
mkt-dup $k=10$	4	2
total	67	28

Table 16: Static ceiling on the TPC-DI conflict cell with 20 numeric and 20 duplicate errors on disjoint keys. Physical enumeration rules out cross-node recovery, and single-node and all-node quarantine both measure 635.627. Computed from d8p-ceiling.json.

plan	NRD	clean collateral
static, dedup@stg_financial	0.596	0
static, dedup@fact_financials	0.596	0
static, quar@fact_financials	635.6	0.0955
static set, dedup@stg + dedup@fact	0.596	0
static set, dedup@stg + quar@fact	635.3	0.0955
static set, quar@stg + dedup@fact	635.6	0.0955
conditional @stg_financial	0.470	0

12.4 Response classes and the location boundary

Classes are distinct measured responses within a cell, and counting by exact damage signature, the mismatch multiset per sink, gives the same partition everywhere.

Twenty-nine pairs of placements share a disposition and differ only in node. The 17 whose nodes both lie on the fork carrying the error measure identically to every recorded digit. Quarantine at stg_financial equals quarantine at fact_financials on all six fin-num and fin-dup cells, and the same holds for dedup on the three fin-dup cells, for the conditional policy on the mixed cell, and for the market-fork pair. The remaining 12 pairs separate, and each sets an on-fork node against quarantine at dim_company or at stg_daily_market, where the disposition has nothing to act on and the response is the no-validation anchor at 1.000. Location therefore decides whether the disposition’s effect is obtained, while which effect it is remains the disposition’s choice. This is the F_1 boundary condition the main text reports for this pipeline.

Table 17: Temporal split of the TPC-DI batch at scale factor 3. Dimension sources are shared and not shown. Computed from split-manifest.json.

source	cut	train	validation
FINWIRE FIN	$\text{fin_year} \leq 2010$	75,950	22,626
DailyMarket	$\leq 2017-02-11$	1,025,004	257,764
Trade	$\leq 2016-08-03$	312,782	78,196

12.5 Sink kind and the quarantine ratio at scale

Decomposing the numeric family by sink kind explains the results. Row-preserving sinks measure 0.500003, 0.500025, and 0.500254 at $k=1, 10, 100$, matching the pre-registered relation

$$q(k) = \frac{N}{2N - k}$$

for $N=98,576$. Aggregate sinks measure 1.000, 0.850, and 0.935, and the corresponding macro values are 0.838, 0.737, and 0.795.

12.6 Cost catalog

Measured under the Jaffle convention, single-threaded with warm-up plus five trials, per-op medians, and the worst case over dispositions, policy cost per node ranges from 3.78s at dim_company to 6.39s at fact_market_history, dominated by the dbt deployment path. Detection is 0.04 to 2.3% of policy cost, inverting the Jaffle finding.

13 E7 THE GATED TPC-DI ROUND

13.1 Temporal train and validation split

The TPC-DI generator has no seed control. DIGen 1.1.0 accepts only an output directory and a scale factor, and changing the seed element of the PDGF schema, in both the annotated configuration and the stripped one the generator actually loads, leaves every generated file byte-identical, checked by digest on DailyMarket and on the FINWIRE quarters. Raising the scale factor changes the scale, never the draw. The two snapshots are therefore a temporal split of the single generated batch, and the full-batch anchor on which the E6 mechanism results were measured is untouched.

Each date-bearing fact source is ordered by its own time column and cut at the 0.8 quantile of its row distribution, on a time-value boundary so that no time value appears on both sides. Dimension sources are shared, namely FINWIRE company and security records, customer and account actions, and the industry reference. This is the Jaffle arrangement, whose two snapshots likewise share the customer and product dimensions and differ in history.

13.2 Development measurements and the response table

The E6 measurements were re-run against the train side alone, 82 rows with zero technical failures. Because $q(k)$

Table 18: E7 roster variants, with n the campaign count. Predicted improvement uses development data alone. Computed from p5-precheck-result.json.

roster	n	policy AURD	predicted	observed
conflict-heavy	18	0.6292	10.96% LAUNCH	10.26%
+ controls	20	0.6663	9.47% REFUSED	—
balanced	18	0.6389	7.80% REFUSED	7.77%
+ controls	20	0.6750	6.72% REFUSED	—

makes each injection cardinality a distinct response class, the response table is keyed by family, k , node, and disposition, giving 42 keyed singletons, 67 composed signatures, and zero measurement conflicts, so C5 is computed from the data. Collapsing the key over k would make the verdict depend on shard row order, since the last-written magnitude survives, while under the corrected keying, reversing or shuffling the rows reproduces every digit. Detector bands are frozen from clean train data at twice the observed maximum and read back by the runner, so no band is fitted to the validation snapshot. The cost catalog was re-measured on the same side, the four planning nodes span 6.39 to 8.80s, and the budget grid derived from it makes the feasible set one node, then two, then four.

13.3 Roster variants and the launch rule

All four variants are scored in one sealed artifact before any fresh input exists. The launch rule is mechanical, run the roster the precheck admits. Exactly one qualifies, so no discretion remains, and the refused variants are reported here rather than discarded. The no-win control variants add one FK-orphan and one row-deletion campaign, which enter the mean as exact zeros.

13.4 Sealed protocol and gate outcome

Eighteen campaigns run on the validation snapshot, four financial numeric at $k=1, 100$, two financial duplicate at $k=1, 100$, two null, eight conflict campaigns injecting 20 numeric and 20 duplicate errors on disjoint keys, and one each of market numeric and market duplicate. Targets are hash-ranked under a salt bound to the frozen plans digest and asserted disjoint from every reconstructed development target, and the exploratory round’s targets are additionally asserted disjoint from this round’s, so the two share no injected row. Seven distinct physical plans were executed, each as a dirty-protected and a clean-counterfactual branch, over 126 placements plus 18 anchors, with zero technical failures. Launch criteria C1 to C5 held with a predicted 10.96%, promotion criteria G1 to G10 held, and the gate’s recomputation agrees with the independent aggregator to 10^{-9} . Observed values are mean AURD 0.6577 for the policy against 0.7329 for the strongest static plan, a relative improvement of 10.257%, a bootstrap 95% CI of $[-0.1024, -0.0481]$, an exact sign-flip $p = 2^{-12}$, and 12 of 18 paired differences non-zero at eleven distinct values spanning -0.070 to -0.160 , the

Table 19: E7 per-family mean AURD on the launched roster. The gain is confined to families where several shapes contend at one node, and the rest tie exactly. Computed from p5-summary.json.

family	policy	strongest static	legacy quarantine
fin-mixed	0.606	0.699	95.1
fin-num	0.850	1.000	1894.9
fin-dup	0.110	0.110	2817.9
mkt-dup	0.375	0.375	1.000
fin-null	1.000	1.000	300.7
mkt-num	1.000	1.000	1.000

two $k=1$ numeric campaigns sharing one value. The decision was GO. Computed from p5-protocol.json, p5-plans.json, p5-precheck-result.json, p5-fresh-registry.json, p5-summary.json, and p5-gate-result.json.

13.5 Safety on a natively dirty pipeline

Across the 72 selected placements the policy fired no rule at all on the clean branch and left zero clean collateral. The strongest static plan fired on clean data yet also left zero collateral, because its disposition is exact dedup, whose blast radius is bounded by a key predicate the detector cannot influence, the E3 mechanism. Legacy quarantine is inadmissible on this pipeline, firing on the FK orphans and nulls TPC-DI carries natively, with 0.095 clean collateral and a worst case of $5578\times$. Availability is full everywhere, since no disposition in this action space can fail a build.

13.6 The refused roster, measured

The balanced roster was sealed as an exploratory round with no promotion gate and a forbidden-claims list that bars reporting an effect size from it or lowering the SESOI so that it would have been admitted. The freeze tool enforces this rather than leaving it to discipline, refusing to seal a confirmatory protocol on a roster the precheck rejected. Over 18 campaigns and 140 placements with zero failures it measured a relative improvement of 7.767% against a prediction of 7.798%, with a bootstrap 95% CI of $[-0.0879, -0.0233]$ and $p = 0.0039$. The improvement is real and below the effect size declared worth acting on, so a confirmatory round would have failed G9. Per-budget predictions run 0.007 to 0.032 high in absolute NRD for both methods alike, which is why the relative margin transfers even where the absolute level does not.

14 ENVIRONMENT

DuckDB 1.5.4, dbt-core 1.12.2, Python 3.13, and Ubuntu 20.04 host every study except the surrogate-collapse pilot, which ran on an earlier image with Python 3.12 and Ubuntu 24.04. Damage is deterministic given the snapshot, so the difference affects timing only, and no timing from that pilot is used.

15 ARTIFACT GUIDE

The artifact ships the figure-generation code, the physical harnesses for every study reported here, the evaluation pipeline for all three fresh rounds covering freeze, identifiability precheck, fresh materializer, sharded runner, independent aggregator, and one-shot gate, the TPC-DI loader and dbt project, and the measurement evidence each table above names by filename, with per-file content hashes.

Ordering is enforced by the artifacts themselves. Development measurements precede the protocol seal, the seal precedes fresh materialization, the runner computes no statistics, the aggregator recomputes everything from raw rows, and the gate refuses to run twice.